HOLY ANGELS CONVENT SCHOOL

ACADEMIC YEAR:2024-2025

PROJECT REPORT ON:

SHOP DEALING AND MANAGEMENT

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CLASS : 12^{TH} – A

ROLL NO : 45

Subject : computer science

INTERNAL EXAMINER: **EXTERNAL:**

PRINCIPAL:

CERTIFICATE

This is to certify that **ABULLAH KAZMI**, a student of **Class 12-A**, has successfully completed the project titled **"Shop Dealing and Management"**.

This project was undertaken under the guidance of our computer science teacher **AMIT BHATIA** and CBSE, demonstrating a thorough understanding and application of the principles related to shop dealing and management.

Project Completion Date: [28TH AUGUST.2024]

Issued by:

MR. AMIT BHATIA

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ACKNOWLEDGEMENT

It is with pleasure that I acknowledge my Sincere gratitude to our teacher, ***MR. AMIT BHATIA*** *who taught and Undertook the responsibility of teaching the subject computer science. I have Been greatly benefited from his classes. I am especially indebted to our Principal* ***Sr. Julie*** *who has always been a source of encouragement and support and without whose inspiration this project Would not have been a successful I would Like to place on record heartfelt thanks to her. Finally, I would like to express my Sincere appreciation for all the other Students in my batch their friendship & the fine time that we all shared Together.*

INTRODUCTION

The "Shop Dealing and Management System" is a computerized solution designed to efficiently manage daily shop operations. This project focuses on streamlining the process of handling item inventories, sales, and profit calculations using Python and SQL for database connectivity.

The system keeps detailed records of available stock, sold items, and financial data such as cost price, selling price, and overall profits. By leveraging user-defined functions, loops, and data structures like lists and dictionaries, the project enables efficient data handling and automation of calculations.

This system helps shop owners or managers monitor stock levels, analyze sales trends, and ensure proper inventory management. It simplifies business operations, reduces human error, and provides a reliable, real-time update of stock and sales data, thus enhancing the overall productivity of the shop.

OBJECTIVE OF THE PROJECT

- The objective of this project is to develop a Python-based shop management system using SQL connectivity that allows a shopkeeper to efficiently manage item inventories and sales. The system aims to

1. Simplify Record Management: Enable the shopkeeper to add, update, and maintain records of items available in the shop, including item names, quantities, cost prices. and selling prices.

2. Track Inventory :- Allow real-time tracking of sold items and update stock levels automatically to ensure accurate inventory control

3. Profit Calculation:- Provide a feature to calculate the total profit based on the difference between the cost price and selling price of items sold, helping the shopkeeper analyze financial performance.

4. **User-Friendly Operations** :- implement user-defined functions and basic Python constructs such as loops, lists, and dictionaries to ensure ease of use for the shopkeeper with minimal technical knowledge.

5. **Seamless Data Storage:-** Store all data securely in an SQL database, ensuring the longevity and integrity of records for future reference and updates

- This project aims to streamline shop operations, reduce manual work, and provide a comprehensive solution for managing inventory and finances

ABOUT THE PROJECT

Shop Dealing and Management System

Objective:

The aim of this project is to build a simple shop management system that records. updates, and manages item inventories, keeps track of sold items, and calculates profits. The system uses Python for application logic and SQLite for database connectivity

Project Breakdown (modules):-

1. Database Setup

This section sets up the SQLite database and table to store the items" data

Functionality Creates a table items with columns id, name, quantity.

cost price, and selling price to store details about each item

2. add_item() Adding New Items

This function allows the shopkeeper to add new items to the inventory

This function asks the user for the item name, quantity cost price, and selling price. The details are then stored in the items table. The INSERT SQL command is used to insert the new record into the database.

Example:- input

```
Enter item name: Shoes
Enter quantity: 20
Enter cost price: 500
Enter selling price: 700
```

output :-

```
Item added successfully!
```

3. view_items() – Viewing All Items

This function displays all the items currently available in the inventory.

- **Explanation**:- it fetches all records from the ‘items’ table using the SELECT* query and displays each item’s details in a formatted manner.

Example. Output

```
ID: 1, Name: Shoes, Quantity: 20, Cost Price: 500, Selling Price: 700
ID: 2, Name: Bags, Quantity: 15, Cost Price: 300, Selling Price: 500
```

4 . sell_items()-Selling Items and Updating Stock

This function manages sales and updates the item quantities

- **Explanation :-** The shopkeeper enters the item ID and quantity sold. The system checks if the available stock is sufficient, then updates the quantity of the item in the database

- **Example**:- Input

```
Enter the item ID to sell: 1
Enter the quantity sold: 5
```

Output:-

```
Sold 5 items. Stock updated.
```

5. calculate profit()-Calculating Total Profit

This function calculates the profit based on the difference between the selling price and cost price of items.

- **Explanation** The profit for each item is calculated as (selling price- cost price) quantity. The total profit is then displayed
- **Example:-** output

```
Total profit: 4000
```

Main Menu:

This is the main control structure of the program that provides the shopkeeper with a list of options to choose from, using a while loop to keep the system running until the user exits.

```
while True:
    print("\n--- Shop Management System ---")
    print("1. Add New Item")
    print("2. View All Items")
    print("3. Sell Item")
    print("4. Calculate Total Profit")
    print("5. Exit")
```

- **Explanation**: This function presents a menu to the shopkeeper, allowing them to select various functionalities. Based on the user's input, the corresponding function is executed.

Conclusion:

This project module provides a comprehensive approach to managing a shop's inventory, sales, and profit calculations using Python and SQLite. By utilizing user-defined functions, loops, and database queries, the system offers a user-friendly experience to help shopkeepers efficiently manage their stock

FRONTEND AND BACKEND

MySQL - The Relational Database Management System

MySQL serves as the backend database for the Student Management System, storing and managing student-related information. The structured schema ensures efficient storage, retrieval, and manipulation of data.

Python - The Programming Language Python

With its versatility and readability, powers the logic and user interface of the Student Management System. The script employs Python's database connectivity features to seamlessly integrate with MySQL, providing a cohesive and interactive experience for users and administrators

SIGNIFICANCE OF THE PROJECT

- **This Shop Dealing and Management System project plays a crucial role in streamlining small to medium-sized businesses by offering an efficient and user-friendly way to manage inventory and financial transactions. The system allows shopkeepers to record item details, track sales, monitor stock levels, and calculate profits, all within a simple, console-based interface.**

- **The project's significance lies in its ability to reduce manual errors, which are common in traditional pen-and-paper methods, and provide real-time data on stock availability and profitability. With features such as automated stock updates and profit calculations, the shopkeeper can focus more on customer service and growth rather than managing tedious tasks. The system also provides a structured and organized database for storing and retrieving information, ensuring that inventory data remains accessible, accurate, and secure**

- **By using Python and SQL, this project offers an introduction to real-world applications of programming and database management, making it highly relevant for students and professionals in the field of software development. Additionally, the project is scalable and can be expanded to include features such as sales reports, customer data, and integration with other accounting tools, offering long-term benefits to businesses Overall, this project represents a practical solution to inventory and shop management challenges, contributing to business efficiency and operational effectiveness.**

FLOWCHART